

Relativistic Time Signal Math: Lorentz Composition, Dynamic Auto-Scaling Bounds, and Phase Orbit Space Mapping

Time Dilation DAW Research Group — Version 0.0.1

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1 Overview

In Time Dilation DAW, time signals ($\gamma(t)$ velocity fields and local coordinate time t_{local}) operate as native audio-rate control streams. Objects such as `time.math~`, `time.scope`, and `time.xy` process, combine, and visualize relativistic time dilation vectors. This paper formalizes the Lorentz composition law for time signals and the dynamic auto-scaling algorithms for 2D phase visualizers.

2 Lorentz Time Signal Composition Law

When two independent relativistic time dilation sources with factors $\gamma_1 \geq 0.0$ and $\gamma_2 \geq 0.0$ are combined via relativistic addition (`time.math~` mode 0), their effective normalized velocities v_1 and v_2 relative to stationary coordinate time are:

$$v_1 = \gamma_1 - 1.0, \quad v_2 = \gamma_2 - 1.0 \quad (1)$$

By Einstein's Special Theory of Relativity, velocity addition in a spacetime manifold with speed-of-light bound $c = 4.0$ follows:

$$v_{\text{comp}} = \frac{v_1 + v_2}{1.0 + \frac{v_1 \cdot v_2}{c^2}} \quad (2)$$

The composite relativistic time dilation factor γ_{comp} is derived as:

$$\gamma_{\text{comp}} = 1.0 + v_{\text{comp}} = 1.0 + \frac{(\gamma_1 - 1.0) + (\gamma_2 - 1.0)}{1.0 + \frac{(\gamma_1 - 1.0)(\gamma_2 - 1.0)}{c^2}} \quad (3)$$

This formula guarantees:

1. **Causality Bound:** $v_{\text{comp}} < c$ for all finite $v_1, v_2 < c$.
2. **Identity Element:** Combining any time signal γ_1 with un-dilated time $\gamma_2 = 1.0$ yields $\gamma_{\text{comp}} = \gamma_1$.
3. **Symmetry:** $\gamma_{\text{comp}}(\gamma_1, \gamma_2) = \gamma_{\text{comp}}(\gamma_2, \gamma_1)$.

3 Dynamic Auto-Scaling Bounds Algorithms

3.1 Continuous Telemetry Scope Auto-Scaling (`time.scope`)

For a time signal sequence $y[i]$ over a sliding window history $i \in [0, K - 1]$, the peak amplitude M is tracked:

$$M = \max_{0 \leq i < K} |y[i]| \quad (4)$$

To prevent waveform clipping or off-screen rendering when γ scales high ($\gamma \gg 1.0$), the scope peak scale $S[n]$ is smoothed via exponential moving average:

$$S[n] = S[n - 1] + \lambda \cdot (\max(1.0, M \cdot 1.15) - S[n - 1]) \quad (5)$$

where $\lambda = 0.1$ is the smoothing coefficient and 1.15 provides a 15% visual headroom margin.

3.2 Dual-Time 2D Phase Lissajous Orbital Bounds (`time.xy`)

For dual time signals $(x[i], y[i])$, the phase orbit radius $r[i]$ is:

$$r[i] = \sqrt{x[i]^2 + y[i]^2} \quad (6)$$

The maximum radial extent $R_{\max}$ over the orbital window is:

$$R_{\max} = \max_{0 \leq i < K} r[i] \quad (7)$$

The 2D visual radius $R_{\text{scale}}[n]$ auto-scales smoothly to enclose the phase orbit:

$$R_{\text{scale}}[n] = R_{\text{scale}}[n - 1] + \lambda \cdot (\max(1.0, R_{\max} \cdot 1.15) - R_{\text{scale}}[n - 1]) \quad (8)$$

Normalized coordinates plotted on the screen vector canvas are $(\hat{x}[i], \hat{y}[i]) = (x[i]/R_{\text{scale}}, y[i]/R_{\text{scale}})$, ensuring full orbital visibility at any dilation magnitude.